In many sectors in Mathematics, the 'Invariant' objects are quite important.

[Def. Let K/F be a finite separable extension, and let L/F be Galois closure. Define

$$N_{K/F}: K \to F;\ \alpha \mapsto \prod_{\sigma \in Emb_F(K,L)} \sigma(\alpha)$$

$$Tr_{K/F}: K \to F;\ \alpha \mapsto \sum_{\sigma \in Emb_F(K,L)} \sigma(\alpha).$$

In particular, if K/F is Galois, then $L=K$, so that $Emb_F(K/L) = Gal(K/F)$.

[Prop.

1). $N_{K/F}$ and $Tr_{K/F}$ are well-defined. (That is, for any $\alpha \in K$, $N_{K/F}(\alpha), Tr_{K/F}(\alpha) \in F$)

2). The $N_{K/F}$ preserves multiplication and the $Tr_{K/F}$ preserves addition.

Proof. Let

[Def. Let E/F be a finite extension of a field F.
Given $\alpha \in E$, a map $m_\alpha : E \to E : x \mapsto \alpha x$ is F-linear map.
Define the Norm of E over F is

$$N_{E/F} : E \to F : \alpha \mapsto \det m_\alpha$$

and the Trace of E over F is

$$Tr_{E/F} : E \to F : \alpha \mapsto tr\, m_\alpha.$$

Note: Tr is an additive Homomorphism on E over F, (also linear functional)
and Norm is a multiplicative Homomorphism on $E^\times$ to $F^\times$

[Prop. Let E/F be a finite extension, and let $\alpha \in E$.
Let $P(x) \in F[x]$ be the minimal polynomial of α over F.
Put $d = \deg P = [F(\alpha) : F]$ and $S = [E : F(\alpha)]$.
Suppose that P factors over $\overline{F}$ as

$$P = \prod_{i=1}^{d} (x - \alpha_i) \quad \text{where } \alpha_1, \cdots, \alpha_d \in \overline{F}$$

Then, the characteristic polynomial of m_α is P^S, and particularly

$$N_{E/F}(\alpha) = \prod_{i=1}^{d} \alpha_i^S \quad \text{and} \quad Tr_{E/F}(\alpha) = S \cdot \sum_{i=1}^{d} \alpha_i.$$

Proof. Let $\{\beta_1, \cdots, \beta_S\}$ be a basis for E over $F(\alpha)$. Since $\{\alpha^0, \cdots, \alpha^{d-1}\}$ is a basis for $F(\alpha)$ over F, $\mathcal{B} = \{\beta_i \cdot \alpha^j \mid 1 \le i \le S,\ 0 \le j \le d-1\}$ is a basis for E over F.